Drone VSLAM

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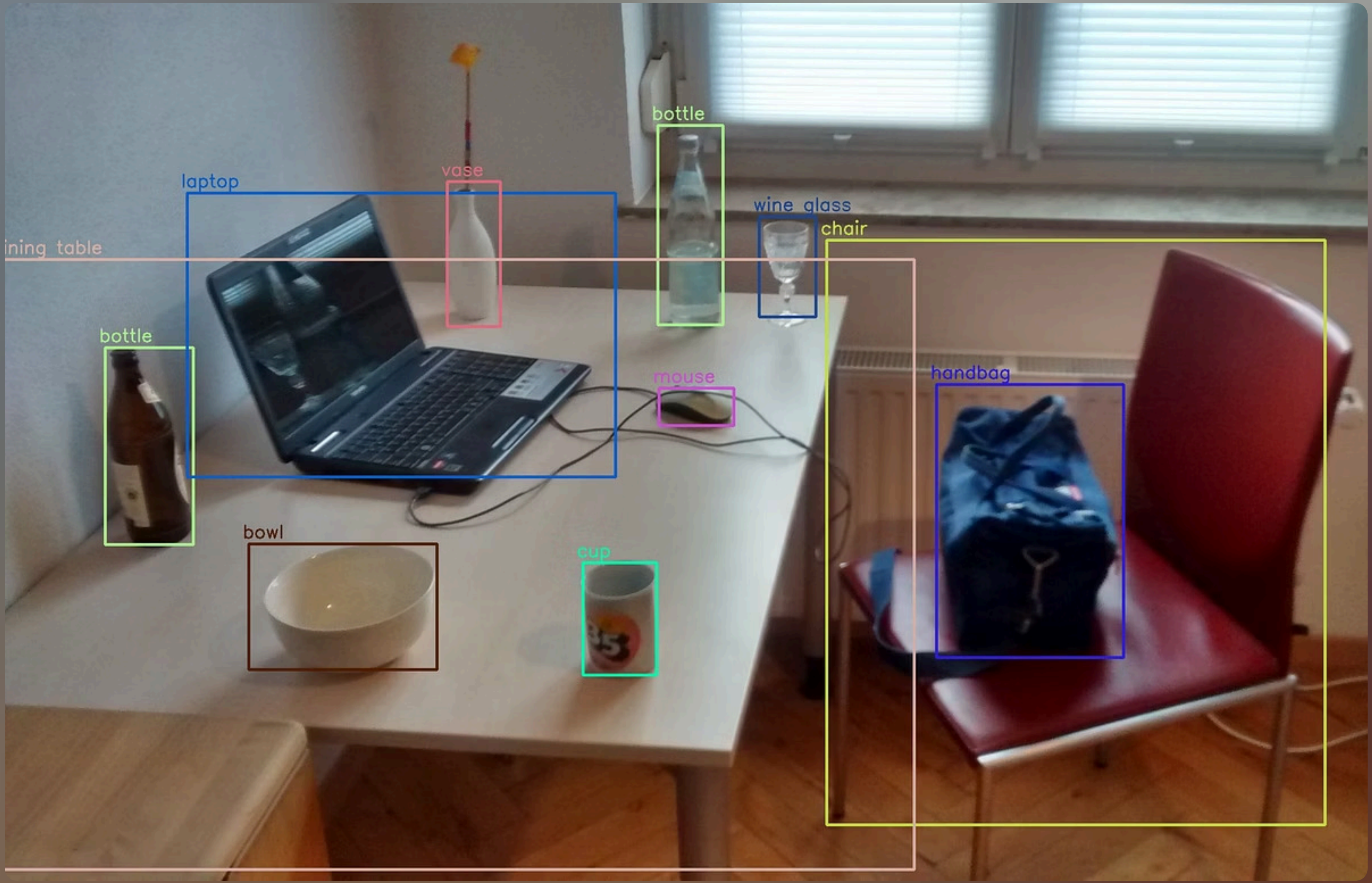
Project Goal

Markerless VSLAM (Visual Simultaneously Localization And Mapping)

10 drones

3D Model



Object Detection

One Project, Two approaches

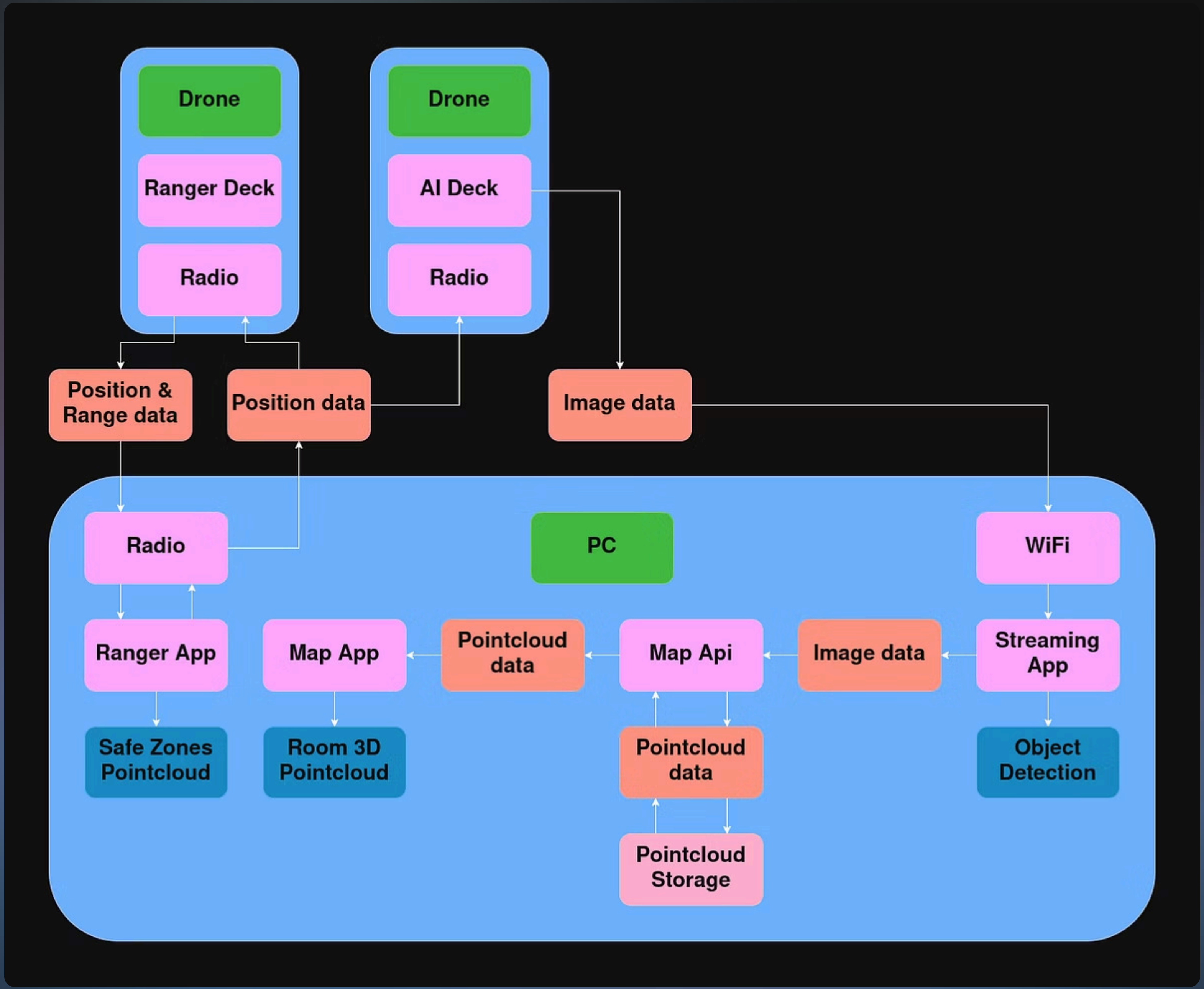
Sequential approach

What drones do we have?

2 drones with a Multiranger deck + flow deck

2 drones with a AI deck + flow deck

What do the drones have to do?



The Sequential Demo

YouTube

Howest CTAI, Industry Project, demo

Master Slave Drone Approach

Swarm Observation Pipeline

Fixed formation, Ranger-only navigation, passive AI-deck streaming.

1. Ranger Safety Check

X_FRONT reads front / left / right / up

2. Formation Step

If clear: X_FRONT + AI drones move 10 cm

3. Blocked → Re-slot

AI core holds; Ranger moves to new front slot

4. Continue + Stream

AI streams stay ON; heading/yaw roles update

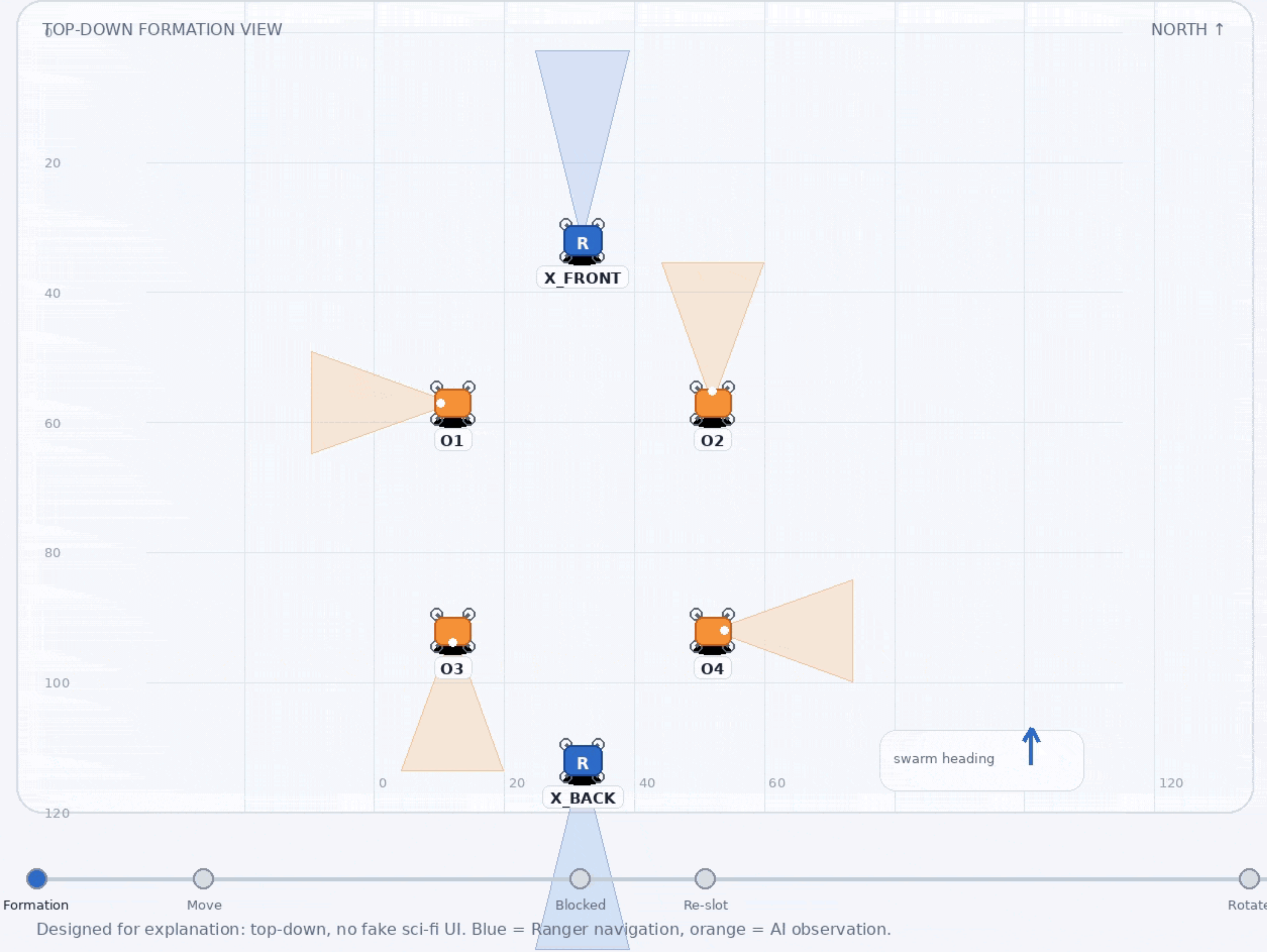
Demo flow clear → move forward → wall detected → AI HOLD → Ranger re-slot → continue EAST

Rule: AI decks observe only. Ranger decks navigate. Unknown space = unsafe.

Full 6-drone fixed formation — simple visual explainer

Ranger drones navigate. AI drones only observe. When front is blocked, Rangers re-slot and the group turns.

MVP behavior: fixed AI core + Ranger re-slot



What is happening?

Full fixed formation

- 2 Ranger drones navigate
- 4 AI drones observe

Navigation

Only X_FRONT / X_BACK Ranger data controls movement.

Observation

O1–O4 stream continuously. No AI navigation.

Safety rule

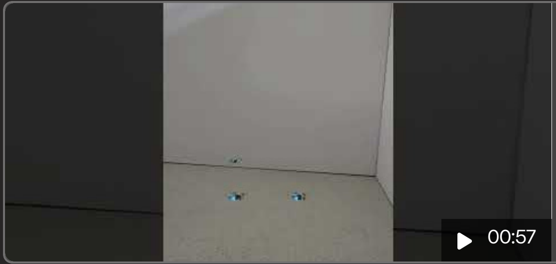
If space is unknown, the formation does not enter.

Turn behavior

AI core holds. Rangers move to new front/back slots.

Formation — Move — Blocked — Re-slot — Rotate — Continue

Designed for explanation: top-down, no fake sci-fi UI. Blue = Ranger navigation, orange = AI observation.



YouTube

DroneDemoVideo

Futher improvement

Add shared localization

Add safety sensors

Add map-based planning

Add better obstacle detection

Improve stability

Add strict emergency

Issues Found

Flow Deck Positioning

Accuracy

Formation Control

Hard coded distance

Drone Stability

Propeller

airflow/downwash

Sensor Limitations

Ranger deck

AI deck

Battery

Point cloud 3D mapping

Mapping Pipeline

Raw image data to world-space position in four stages.

Image Padding

Normalize raw input image

I2P

Extract depth data

Localization / L2W

Transform to world space

Return Position

Output final world-space position

Desired Result vs. Current Implementation

What We Built

- Sequential frame stitching
- Multiple cameras management
- Position tracking
- Loop detection
- API containerization

What's Next

- Separate drone map merging
- Scale-out compatibility
- Error correction

Ease of Use

The Mapping API is fully dockerized — no complex dependencies, no manual configuration. Get up and running in seconds with a single command.

Fully Dockerized

Consistent environments across dev, staging, and production.

Instant Setup

Pre-configured and ready to run — no manual steps needed.

Portable & Scalable

Deploy anywhere — local, cloud, or orchestrated clusters.

One Command to Launch

```
docker compose up
```

Map Merge Bottlenecks

Two main approaches to map merging — both face significant challenges:

Common Map Approach

All agents contribute to a single shared map in real time.

- **Computational overhead:** L2W weight increases *linearly* with each step, making merging progressively slower
- **Depth scale mismatch:** Depth estimation is not metric-scale, making this approach incompatible

Separate Maps Approach

Each agent builds its own map independently, merged afterward.

- **More promising** - avoids real-time merge overhead
- **Depth scale inconsistency:** Irregular depth scales across agents cause misalignment during merging
- **Existing pipelines tested:** Differing depth scales still break the merge

Repositories We Tried

[Covins]

Strengths: Designed for collaborative multi-agent SLAM with a structured server-client architecture.

Limitations: Requires precise depth input via stereo (binocular) cameras or LiDAR - incompatible with monocular depth estimation pipelines.

[Swarm-SLAM]

Strengths: Most mature multi-camera mapping solution; excellent loop closure detection and drift correction over time.

Limitations: Requires stereo/RGBD cameras or LiDAR, plus a high-frequency IMU. Not compatible with monocular setups.

Repositories We Tried

RTAB-Map

Strengths: Gaussian splat-based mapping with built-in depth estimation; produces high-quality, near-complete maps without external depth sensors.

Limitations: Extremely RAM-intensive and computationally heavier than point cloud approaches - not suitable for resource-constrained swarm deployments.

SLAM3R

Strengths: Strong monocular image-to-point-cloud (I2P) pipeline; functional local-to-world (L2W) estimation using sequential frame differencing and landmark detection. Adapted by us for real-time sequential processing.

Limitations: Lacks metric-scale depth accuracy, causing scale inconsistencies across multiple cameras. Requires high frame rates and sharp imagery to maintain tracking stability.

Updates to be Made

Service Separation

Proposal: Decouple individual map processing from the map merging service into independent, separately scalable components.

Impact: Map processing instances can be scaled horizontally across swarm nodes without being constrained by the merging pipeline, improving throughput and resource efficiency in distributed deployments.

Hardware Upgrade

Proposal: Replace monocular cameras with binocular (stereo) camera setups across the swarm.

Impact: Stereo vision provides native metric-scale depth, eliminating cross-camera scale inconsistencies and removing the need for computationally expensive depth estimation - significantly reducing per-node processing load.

YouTube

Drone footage

YouTube

I2P pcds

YouTube

render frames

Detection & classifiction models

Improvements since last presentation

Posture classification

Detector retraining

Real time pipeline

Pipeline

1. AI deck
2. Detection model
3. Classification model
4. Final result

Detection model performance

Precision

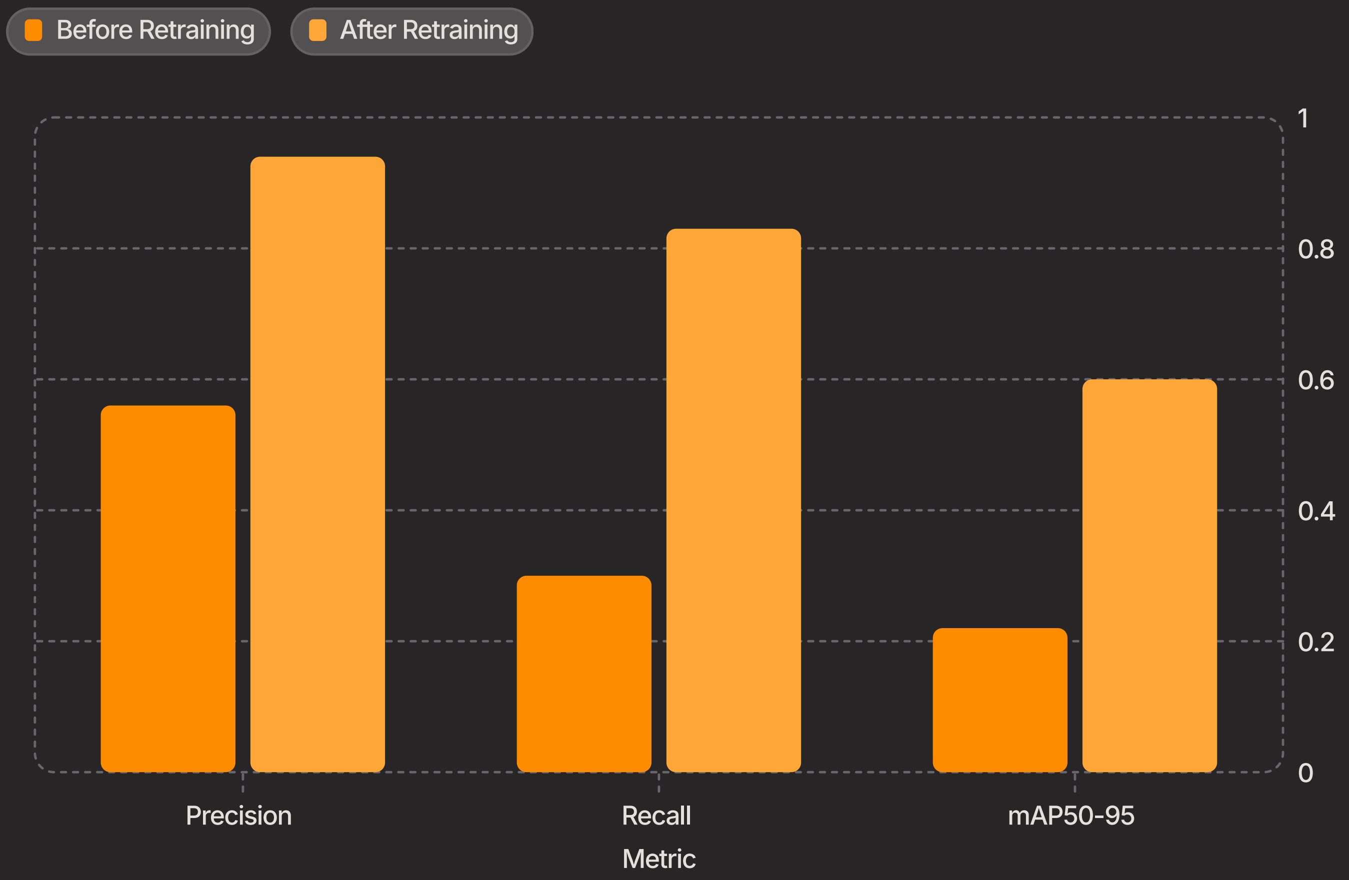
0.56 - 0.94, +38% improvement

Recall

0.30 - 0.83, +53% improvement

mAP50-95

0.22 - 0.60, +38% improvement



Classification model performance

Precision

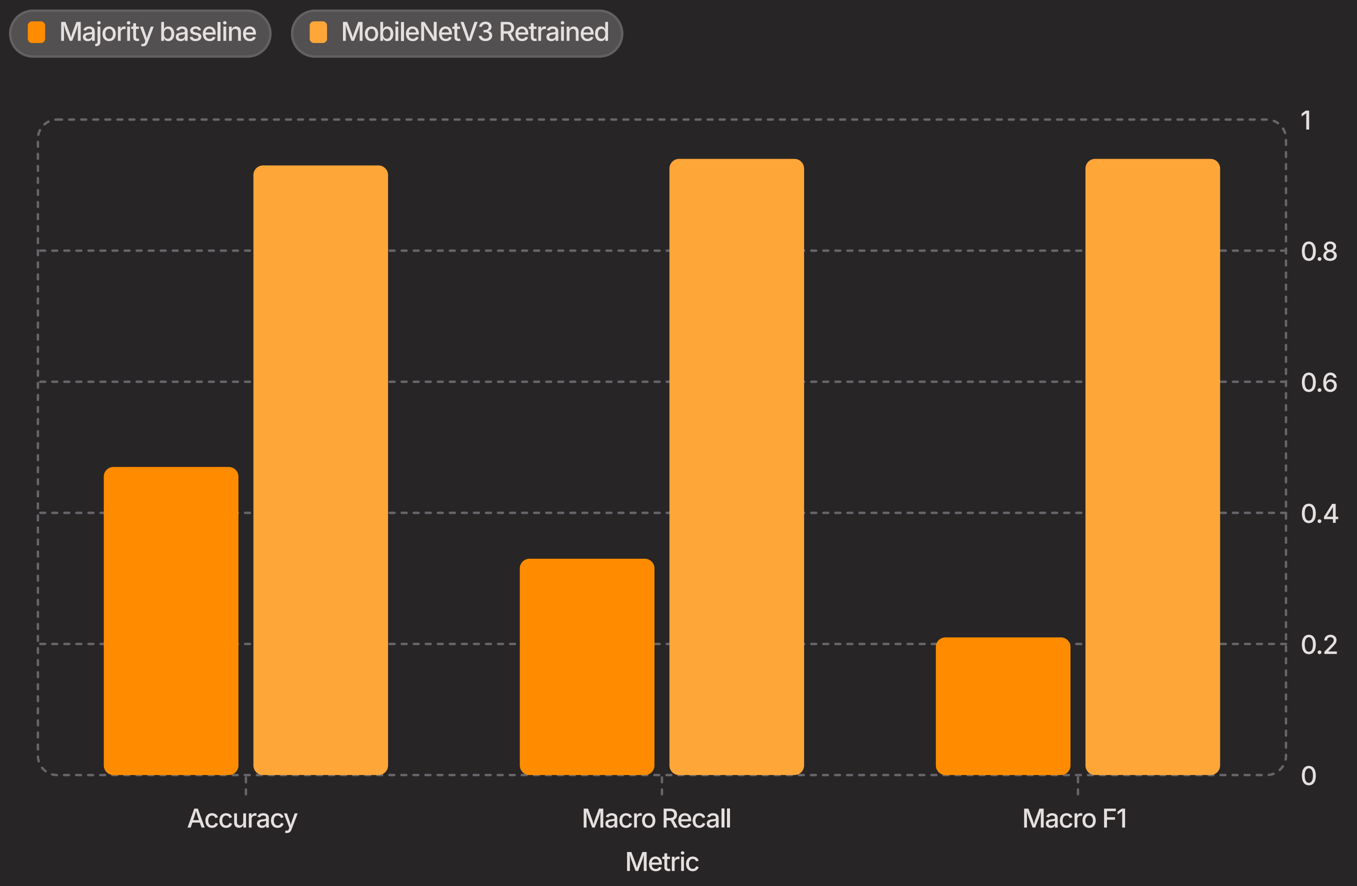
0.47 - 0.93, +46% improvement

Macro Recall

0.33 - 0.94, +61% improvement

Macro F1

0.21 - 0.94, +73% improvement



Confidence thresholds

Human detection

Threshold: **0.23**

Prioritizes recall on noisy drone images. Better to detect extra candidates than miss a person

Posture classification

<50% → person_unkown

Prevents weak crops from being forced into determined class

Limitations

Input quality

Low resolution, blur, partial people

Data collection cost

Manual labeling limits dataset size and diversity

Environment variation

Lighting, room layout, camera angle

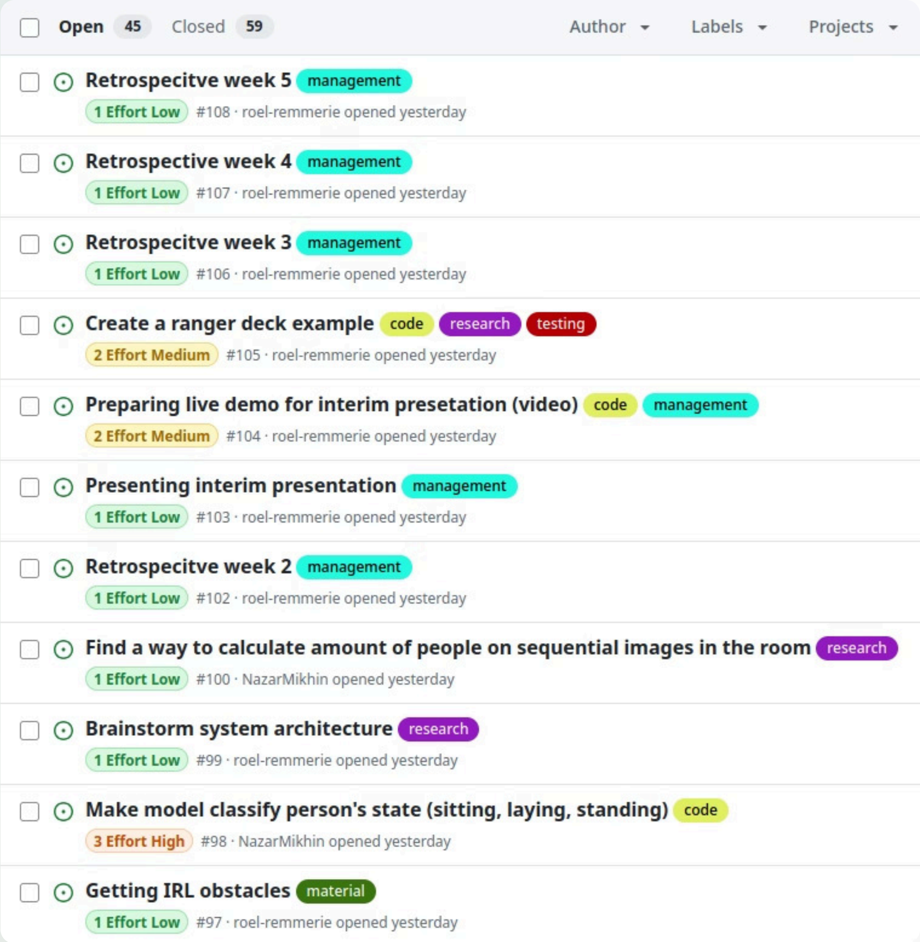
Demostration

YouTube

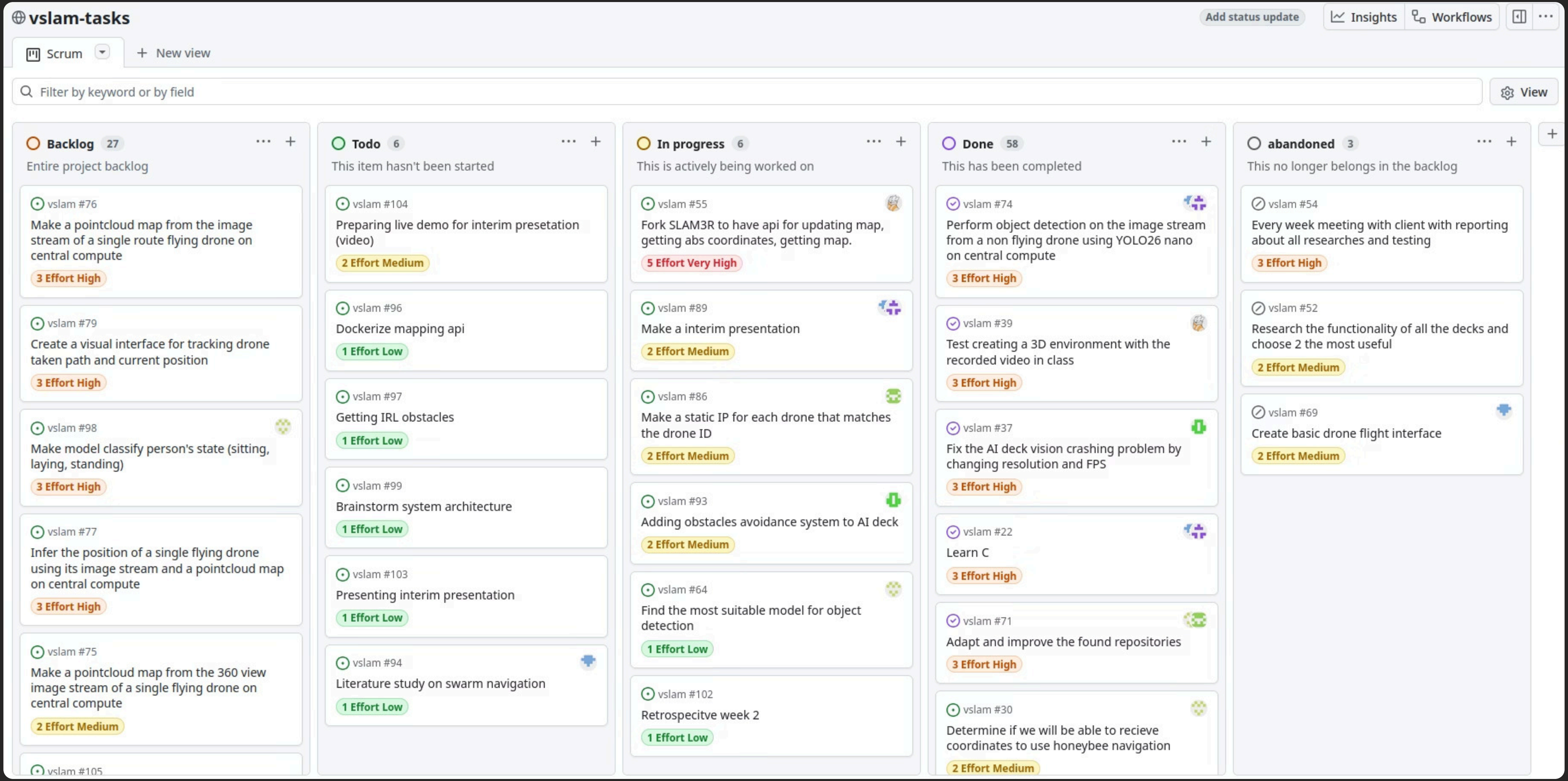
DroneModelDemoVideo

Management

To manage it all



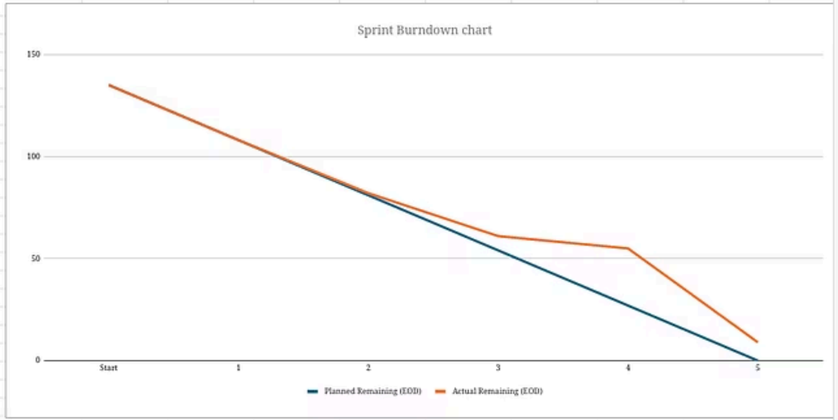
I got 99 issues but a bug ain't one


vslam-tasks
Add status update
Insights
Workflows
Scrum
New view
Filter by keyword or by field
View
Backlog 27
Entire project backlog
vslam #76
Make a pointcloud map from the image stream of a single route flying drone on central compute
3 Effort High
vslam #79
Create a visual interface for tracking drone taken path and current position
3 Effort High
vslam #98
Make model classify person's state (sitting, laying, standing)
3 Effort High
vslam #77
Infer the position of a single flying drone using its image stream and a pointcloud map on central compute
3 Effort High
vslam #75
Make a pointcloud map from the 360 view image stream of a single flying drone on central compute
2 Effort Medium
vslam #105
Todo 6
This item hasn't been started
vslam #104
Preparing live demo for interim presetation (video)
2 Effort Medium
vslam #96
Dockerize mapping api
1 Effort Low
vslam #97
Getting IRL obstacles
1 Effort Low
vslam #99
Brainstorm system architecture
1 Effort Low
vslam #103
Presenting interim presentation
1 Effort Low
vslam #94
Literature study on swarm navigation
1 Effort Low
In progress 6
This is actively being worked on
vslam #55
Fork SLAM3R to have api for updating map, getting abs coordinates, getting map.
5 Effort Very High
vslam #89
Make a interim presentation
2 Effort Medium
vslam #86
Make a static IP for each drone that matches the drone ID
2 Effort Medium
vslam #93
Adding obstacles avoidance system to AI deck
2 Effort Medium
vslam #64
Find the most suitable model for object detection
1 Effort Low
vslam #102
Retrospecitve week 2
1 Effort Low
Done 58
This has been completed
vslam #74
Perform object detection on the image stream from a non flying drone using YOLO26 nano on central compute
3 Effort High
vslam #39
Test creating a 3D environment with the recorded video in class
3 Effort High
vslam #37
Fix the AI deck vision crashing problem by changing resolution and FPS
3 Effort High
vslam #22
Learn C
3 Effort High
vslam #71
Adapt and improve the found repositories
3 Effort High
vslam #30
Determine if we will be able to recieve coordinates to use honeybee navigation
2 Effort Medium
abandoned 3
This no longer belongs in the backlog
vslam #54
Every week meeting with client with reporting about all researches and testing
3 Effort High
vslam #52
Research the functionality of all the decks and choose 2 the most useful
2 Effort Medium
vslam #69
Create basic drone flight interface
2 Effort Medium

Up is Down

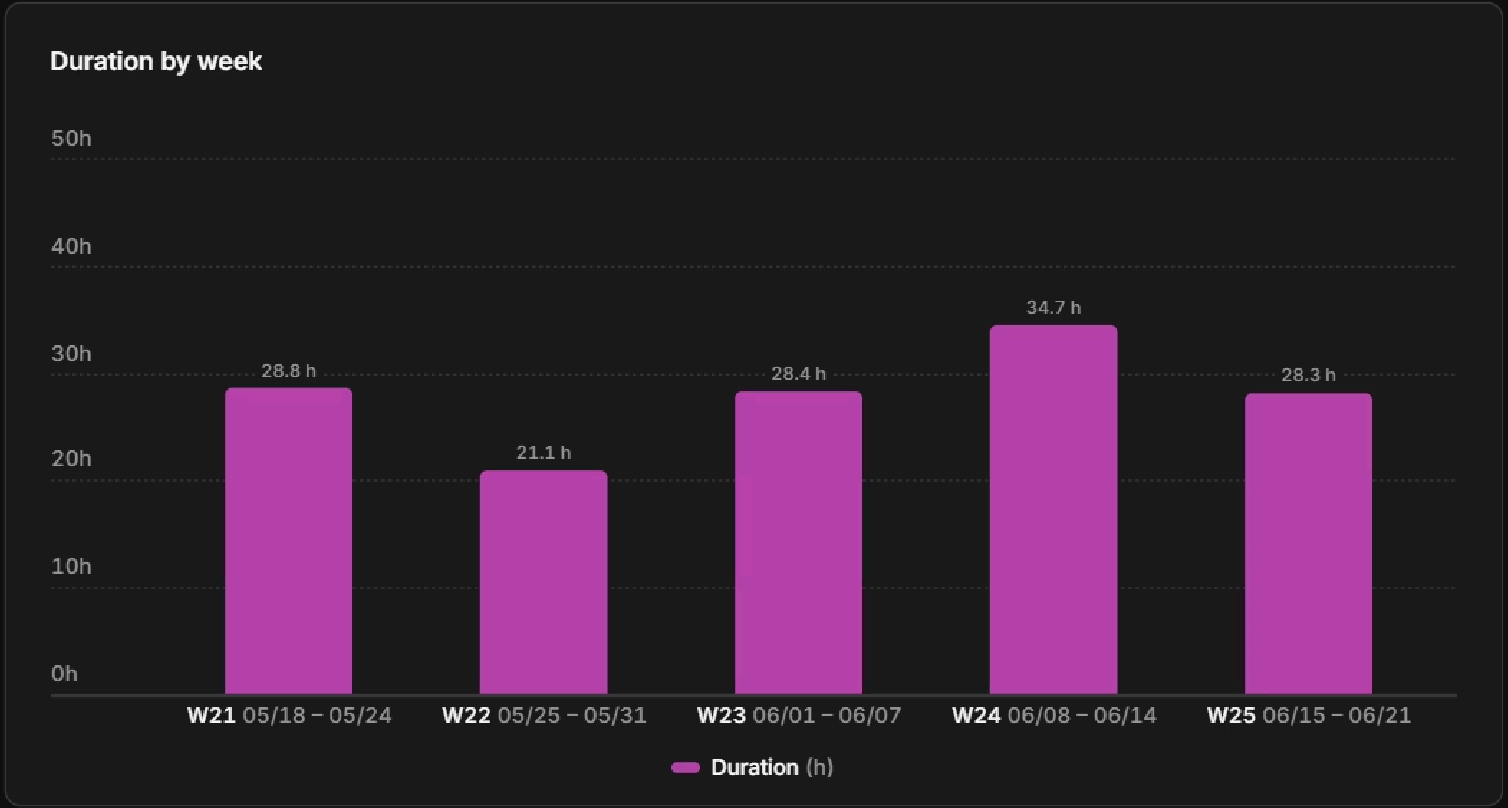
Committed	135
Sprint length (days)	5
Ideal trend (daily)	27

Sprint Day	Planned Remaining (EOI	Actual Delivered (EOI	Actual Remaining (EOD)
Start	135		135
1	108	27	108
2	81	26	82
3	54	21	61
4	27	6	55
5	0	46	9

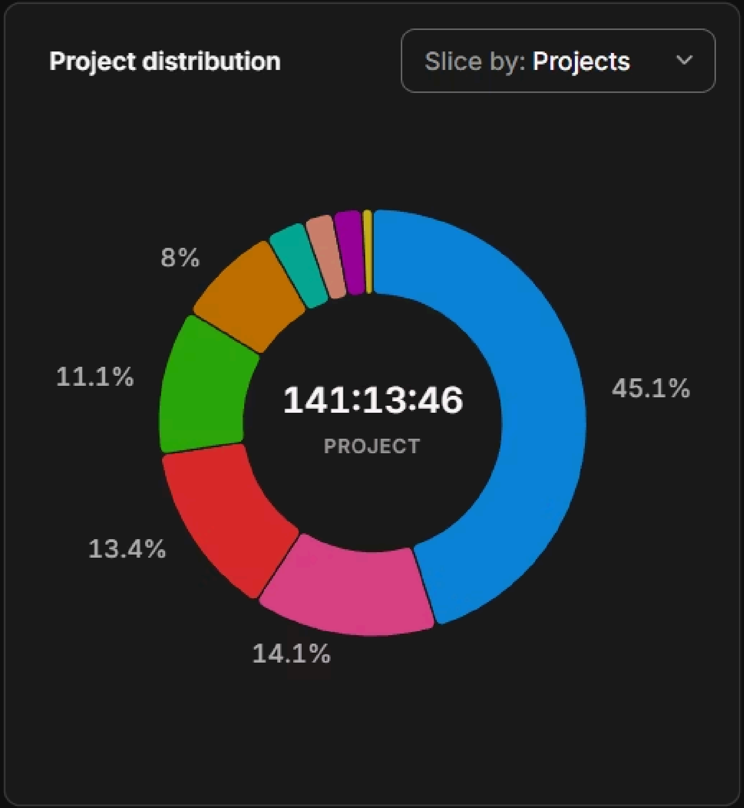


Toggl

Toggl Noah

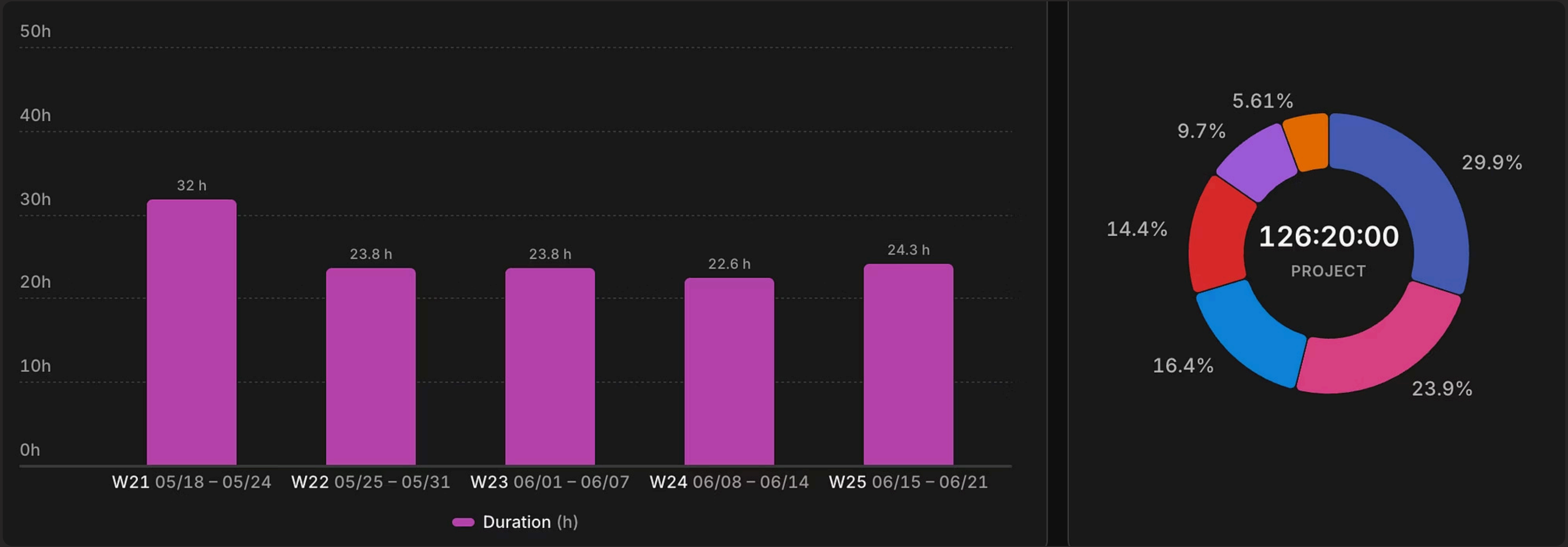


Duration by week
50h
40h
30h
20h
10h
0h
28.8 h
21.1 h
28.4 h
34.7 h
28.3 h
W21 05/18 – 05/24
W22 05/25 – 05/31
W23 06/01 – 06/07
W24 06/08 – 06/14
W25 06/15 – 06/21
Duration (h)

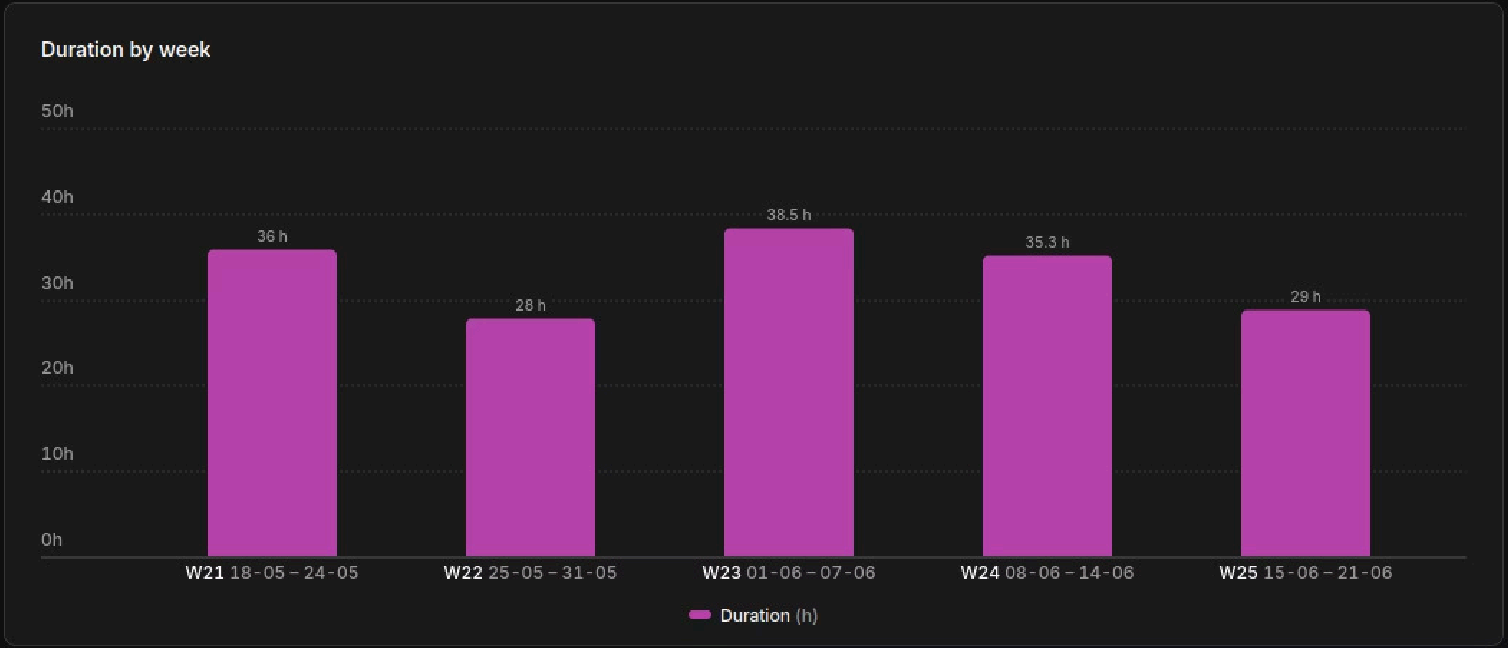


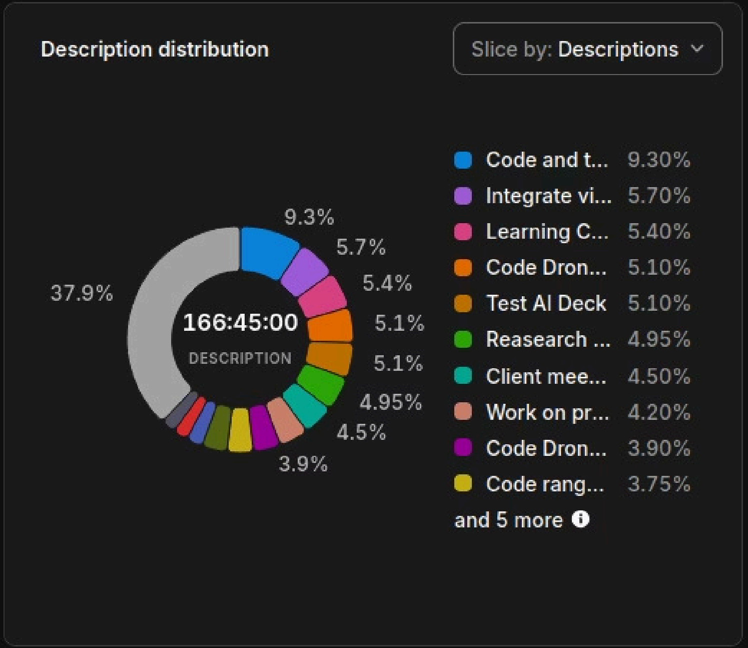
Project distribution
Slice by: Projects
141:13:46
PROJECT
45.1%
14.1%
13.4%
11.1%
8%

Toggl Anton

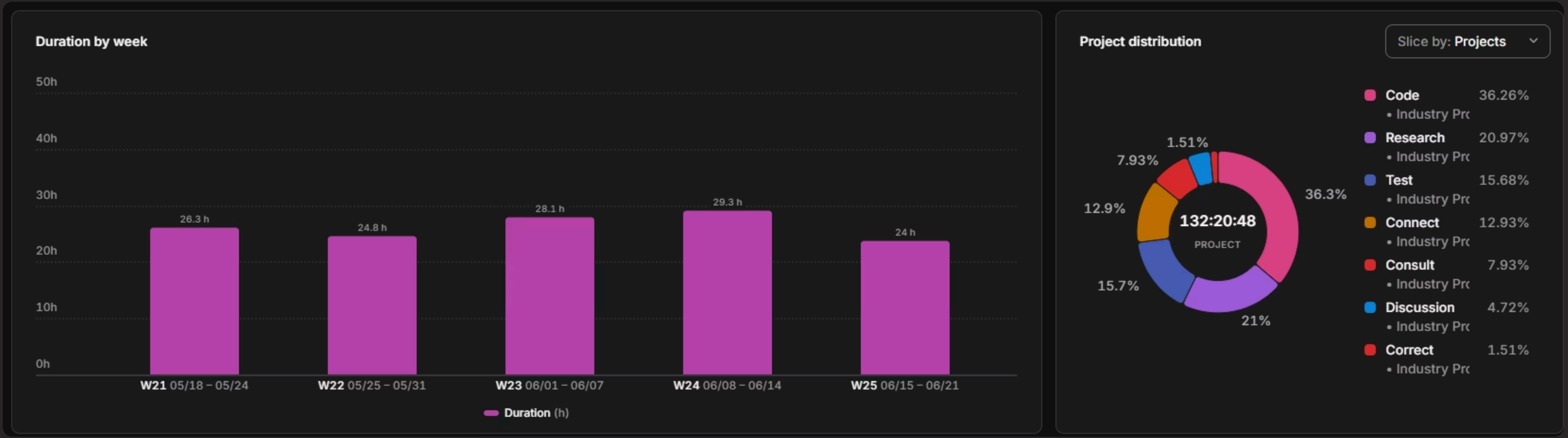

50h
40h
30h
20h
10h
0h
32 h
23.8 h
23.8 h
22.6 h
24.3 h
W21 05/18 – 05/24
W22 05/25 – 05/31
W23 06/01 – 06/07
W24 06/08 – 06/14
W25 06/15 – 06/21
Duration (h)
5.61%
9.7%
29.9%
14.4%
126:20:00
PROJECT
16.4%
23.9%

Toggl Roel

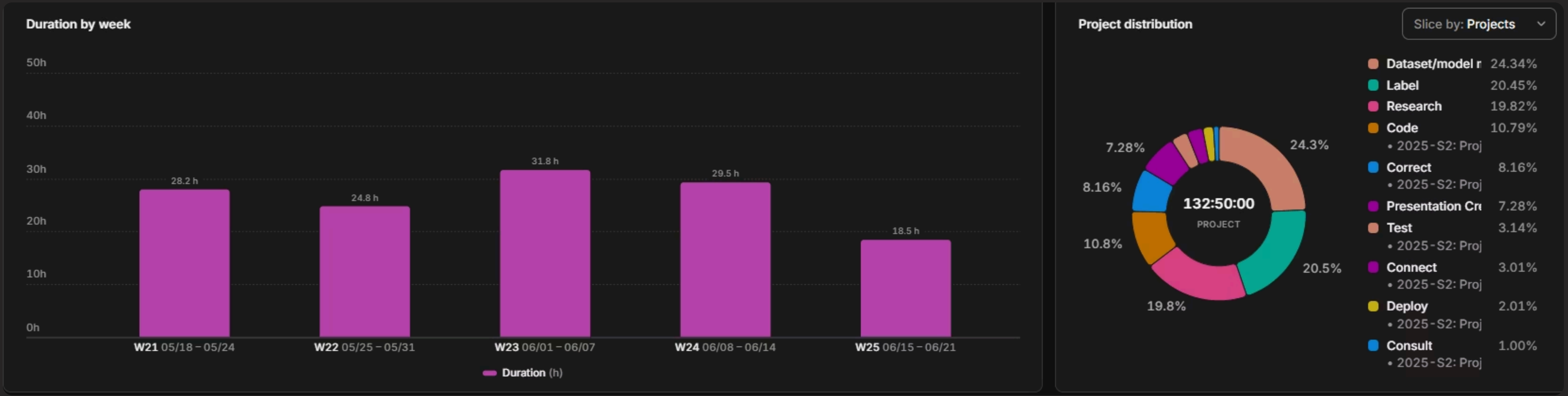

Duration by week
50h
40h
30h
20h
10h
0h
36 h
28 h
38.5 h
35.3 h
29 h
W21 18-05 – 24-05
W22 25-05 – 31-05
W23 01-06 – 07-06
W24 08-06 – 14-06
W25 15-06 – 21-06
Duration (h)


Description distribution
Slice by: Descriptions
37.9%
9.3%
5.7%
5.4%
5.1%
5.1%
4.95%
4.5%
3.9%
166:45:00
DESCRIPTION
Code and t... 9.30%
Integrate vi... 5.70%
Learning C... 5.40%
Code Dron... 5.10%
Test AI Deck 5.10%
Reasearch ... 4.95%
Client mee... 4.50%
Work on pr... 4.20%
Code Dron... 3.90%
Code rang... 3.75%
and 5 more

Toggl Nikita

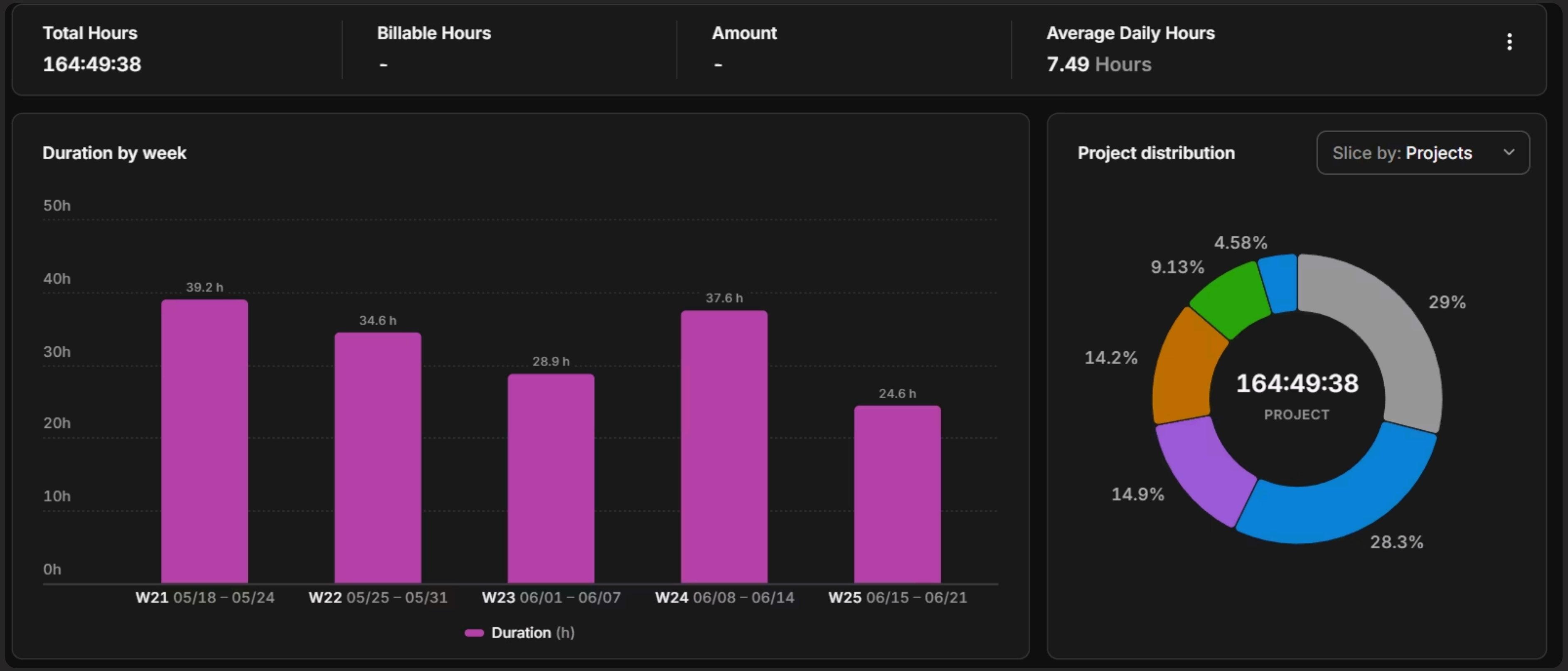


Toggl Nazar



Duration by week
50h
40h
30h
20h
10h
0h
28.2 h
24.8 h
31.8 h
29.5 h
18.5 h
W21 05/18 – 05/24
W22 05/25 – 05/31
W23 06/01 – 06/07
W24 06/08 – 06/14
W25 06/15 – 06/21
Duration (h)
Project distribution
Slice by: Projects
132:50:00
PROJECT
24.3%
20.5%
19.8%
10.8%
8.16%
7.28%
Dataset/model r 24.34%
Label 20.45%
Research 19.82%
Code 10.79%
• 2025-S2: Proj
Correct 8.16%
• 2025-S2: Proj
Presentation Cr 7.28%
Test 3.14%
• 2025-S2: Proj
Connect 3.01%
• 2025-S2: Proj
Deploy 2.01%
• 2025-S2: Proj
Consult 1.00%
• 2025-S2: Proj

Toggl Tofa



Made with GAMMA

Retrospective

Problems and conclusions

- De-motivation due to lack of progress — work must be divided into smaller segments with measurable milestones to avoid burnout.
- Decentralized group structure caused stagnation and duplicated efforts — a dedicated team leader is needed to set direction and track progress.
- Over-reliance on client feedback disrupted sprint goals — external remarks should be evaluated before entering the backlog.
- Client failed to define a clear end result, causing the team to circle without reaching concrete decisions.

Positive sides

- First 3 weeks were highly productive with clear field divisions and easy progress tracking.
- Each team member worked independently and managed their time effectively.
- Initial curiosity and investment led to strong brainstorming and inspiration.
- Excellent client communication — connected weekly with all remarks promptly added to the backlog.
- Dividing working fields between members maximized coverage and output.

We have proof

Project Handover Maak een samenvatting van dit e-mailbericht

RR Roel REMMERIE

Beantwoorden Allen beantwoorden Doorsturen ...

Aan: Bart LEENKNEGT

Do 18-6-2026 18:25

CC: Noah LERNOUT; Nazar MIKHIN; Nikita ILIASHENKO; Anton TYKHONENKO; Timofejs RIMENSONS; Sofie EECKEMAN; Dieter DE PREESTER

Dear Bart,
The full code base (demo, research and tests) is available on github https://github.com/industry-project-vslam/vslam/tree/main
Your many ideas forced us to think bigger. Thank you from the whole team for the opportunity to work on this drone project.

Roel Remmerie - Student 3CTAI

Bachelor Creative technologies and AI
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howest.be/ctai

howest/MULTIMEDIA & CREATIVE TECHNOLOGIES

Beantwoorden Allen beantwoorden Doorsturen

Special thanks

- Paula Acuña Roncancio
- Bart Leenknegt
- All Industry Project coaches